

Pramukh Shankar

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SKILLS

Languages/Tools: Python, R, SQL, SAS, GIS, HTML, STATA, Tableau, SPSS, Excel

Libraries/Frameworks: NumPy, Pandas, Scikit-learn, PostgreSQL, Langchain, Next.js, FastAPI, TensorFlow, PyTorch, Keras

EDUCATION

University of California, Santa Barbara

Goleta, CA

B.S. Statistics & Data Science; B.A. Economics; Minor in Philosophy

Sept. 2021 – June 2025

EXPERIENCE

Research Assistant

Sept. 2024 – June 2025

Health Innovation Lab, UCSB

Goleta, CA

- Built Python data pipelines to analyze 100K+ cancer-trial and exposure records, reducing analysis time by 40%, enabling facility-level exposure scoring across 20+ counties, and surfacing statistically significant exposure-incidence relationships used in public-health risk evaluation.
- Automated analysis of 500+ cross-country drug approvals using SQL and visualization tools, reducing regulatory timeline analysis time by 65% and replacing manual spreadsheet workflows.
- Reduced analysis turnaround time by 60% by automating regression and SQL-based pipelines to screen 200+ chemicals for cancer risk correlations

Research Assistant

June 2023 – Sept. 2023

American Institute for Behavioral Research and Technology

San Diego, CA

- Applied effect size analysis (Cohen's d), z-tests, and Likert-scale modeling to experimental data from 400+ participants to quantify demographic susceptibility to online influence; engineered an R- and Excel-based processing pipeline for CSV and survey data, cutting analysis time by 40%
- Aided in development of APA-submitted manuscript; contributed to literature review, journal selection, and revision workflows
- Built a vote-shift metric and reproducible calculator using Python and SQL, showing how search results can sway up to 20% of votes

Data Intern

June 2022 – Sept. 2022

Mucci Assessment

San Diego, CA

- Developed Python pipeline (NumPy, Pandas) to scrape and clean 500+ pediatrician contacts, reducing manual data entry by 10+ hours per week
- Developed and deployed a backend scheduling application using Python and SQL, reducing manual processing time by 60% and saving 15+ staff hours per week.

PROJECTS

Thesis-Graph LLM-assisted Argument Mapping App

- Developed a full-stack application (Next.js, FastAPI, React Flow) that extracts claims from text and visualizes reasoning as an interactive argument graph with editable nodes and causal links.
- Implemented backend endpoints for node extraction, edge suggestion, and essay composition using RAG and LLM-powered prompt pipelines.
- Developed scalable data storage and authentication systems (SQLite, JWT) and built an interactive React Flow visualization layer supporting autosave, version tracking, and dynamic node-edge graph editing

Run For Office with Nationbuilder

- Built a domain-whitelisted Playwright crawler for county sites nationwide, capturing 1K+ HTML/PDF links per run and replacing Google Search API workflows—cutting data collection time by 80%.
- Integrated PyMuPDF and PDFPlumber to standardize PDF processing, facilitating LLM extraction across diverse document formats and reducing data errors by 15% company-wide
- Designed grounded LLM extraction at the county level for open offices (OpenAI o1), cutting nationwide collection costs by 70% (\$35,000+)

Pokémon Type Prediction & Analysis

- Trained Random Forest, Decision Tree, LDA, and KNN models on 800+ entries with 20+ attributes, achieving ROC_AUC of 0.69
- Achieved 28.2% prediction accuracy, outperforming the 5.6% random baseline by 5×
- Implemented an interactive Tableau dashboard using Kaggle datasets to dynamically visualize and rank Pokémon type matchups

Classifying Tennis Player Styles

- Created a Render-hosted interactive web app that scrapes and analyzes pro tennis match data, classifies player styles from rally length, shot frequency, and winner types, and benchmarks performance against tour averages for real-time insights
- Applied k-means, DBSCAN, and fuzzy c-means clustering to segment player styles from match features

OTHER

Team Leader

May 2025 – Oct. 2025

AmeriCorps NCCC Forest Corps

Sacramento, CA

- Led a team of 7 to support reforestation through invasive species removal, stand improvement, and land mapping

President

Oct. 2021 – June 2025

UCSB Club Tennis

Goleta, CA

- Managed club operations for 50+ students, coordinating fundraisers, social events, competitive practice sessions, travel logistics, meetings, and tryouts